Zander W. Blasingame

zander.blasingame@gmail.com zblasingame.github.io zander-blasingame zblasingame @zwblasingame

Professional Experience

AITHYRA — **Postdoctoral Fellow** — 2026 – Present
Vienna, Austria
Supervisor: Alexander Tong
• Stochastic flow maps, likelihoods for continuous normalizing flows, AI for Science

Clarkson University — **Graduate Research Assistant** — 2018 – 2025
Potsdam, NY
Supervisor: Chen Liu
• Diffusion models, neural differential equations, flow-based generative modeling
• Face morphing attacks, biometrics, anomaly detection

Clarkson University — **Undergraduate Research Assistant** — 2016 – 2018
Potsdam, NY
Supervisor: Chen Liu
• GAN inversion, probability density estimation, zero-day malware detection

Griffiss Institute — **Engineering Intern** — 05.2016 – 08.2016
Rome, NY

UNH InterOperability Lab — **Engineering Intern** — 07.2014 – 08.2014
Durham, NH

Education

Clarkson University
Potsdam, NY

Ph.D. in Electrical and Computer Engineering — 2025
Thesis: On Guided and Reversible Solvers for Neural Differential Equations
M.Sc. in Electrical and Computer Engineering (en route to Ph.D.) — 2025
B.Sc. in Computer Engineering, *with Great Distinction* — 2018

Publications

In reverse chronological order [*]Denotes equal contribution [†]Denotes equal advising

NeurIPS — **Strong Stochastic Flow Maps** — 2026
Oral
*S. McCallum[*], **Z.W. Blasingame**[*], T. Herschell, N. Rindtorff, A. Tong[†], and J. Foster[†]*
Published at NeurIPS (2026) · *Oral [top 0.4% of submissions]*
Presented at ICML 2026 SPIGM workshop
arxiv openreview code pdf slides video

NeurIPS — **Improved Training of Strong Stochastic Flow Maps** — 2026
Workshop
***Z.W. Blasingame**, A. Franca, N. Rindtorff, and A. Tong*
Presented at NeurIPS 2026 STODY workshop
arxiv openreview code pdf

NeurIPS — **Global Transport Couplings for Classifier-Free Guided Flows** — 2026
Workshop
*K. Petrović, **Z.W. Blasingame**, D. Rehman, İ.İ. Ceylan, M.M. Bronstein, S.Y. Zhang[†], L. Atanackovic[†], and A. Tong[†]*
Presented at NeurIPS 2026 STODY workshop
arxiv openreview code pdf

NeurIPS — *Workshop*
Normalizing Flow Maps — 2026
*F.M. Ruscio[*], T.J. Hitge[*], Z. Peng, D. Rehman, **Z.W. Blasingame**, N.M. Boffi, M.M. Bronstein, A. Tong, and J. Bose*
Presented at NeurIPS 2026 STODY workshop
arxiv openreview code website pdf

ICML — *Oral*
Rex: A Family of Reversible Exponential (Stochastic) Runge–Kutta Solvers — 2026
***Z.W. Blasingame** and C. Liu*
Published at ICML (2026) · *Oral [top 0.7% of submissions]*
Presented at ICLR 2026 DeLTa and ReALM-GEN workshops
arxiv openreview code website pdf poster slides

ICLR — *Workshop*
Stochastic Few-step Models — 2026
*R. Passaro, **Z.W. Blasingame**, M.M. Bronstein, and A. Tong*
Presented at ICLR 2026 DeLTa and ReALM-GEN workshops
openreview pdf

NeurIPS
Greed is Good: A Unifying Perspective on Guided Generation — 2025
***Z.W. Blasingame** and C. Liu*
Published at NeurIPS (2025)
Presented at ICML 2025 EXAIT workshop and ICLR 2025 FPI workshop
arxiv openreview pdf poster slides

IEEE ASAP
Scalable Malware Detection Framework Using Performance Counters and Gradient Boosting — 2025
*C. Woralert, C. Liu, and **Z. Blasingame***
Published at IEEE ASAP (2025)

HASP
Feasibility of Scalable Performance Counter Malware Detection Framework Using LightGBM — 2025
*C. Woralert, C. Liu, and **Z. Blasingame***
Published at HASP (2025)

IEEE MOST
Poster: Adapting Pretrained Vision Transformers with LoRA Against Attack Vectors — 2025
*R.E. Neddo, S. Willis, **Z.W. Blasingame**, and C. Liu*
Published at IEEE MOST (2025)

NeurIPS
AdjointDEIS: Efficient Gradients for Diffusion Models — 2024
***Z.W. Blasingame** and C. Liu*
Published at NeurIPS (2024)
Presented at NeurIPS - New Frontiers in Adversarial Machine Learning
arxiv openreview code website pdf poster slides

IEEE S&P
Fast-DiM: Towards Fast Diffusion Morphs — 2024
***Z.W. Blasingame** and C. Liu*
Published at IEEE S&P (2024)
arxiv code website pdf

IJCB — *Spotlight*
Greedy-DiM: Greedy Algorithms for Unreasonably Effective Face Morphs — 2024
***Z.W. Blasingame** and C. Liu*
Published at IJCB (2024) · *Spotlight*
arxiv code website pdf poster

IEEE TBIOM — 2024

Leveraging Diffusion for Strong and High Quality Face Morphing Attacks
Z.W. Blasingame *and C. Liu*
Published at IEEE TBIOM (2024)
Presented at IJCB (Oral)
arxiv code website pdf poster slides

IJCB — 2024

The Impact of Print-Scanning in Heterogeneous Morph Evaluation Scenarios
R.E. Neddo, ***Z.W. Blasingame****, and C. Liu*
Published at IJCB (2024)
arxiv pdf poster

HASP — 2024

Towards Effective Machine Learning Models for Ransomware Detection via Low-Level Hardware Information
C. Woralert, C. Liu, and ***Z. Blasingame***
Published at HASP (2024)

IEEE TCAS-I — 2023

HARD-Lite: A Lightweight Hardware Anomaly Realtime Detection Framework Targeting Ransomware
C. Woralert, C. Liu, and ***Z. Blasingame***
Published at IEEE TCAS-I (2023)

AsianHOST — 2023
Oral

A Comparison of One-class and Two-class Models for Ransomware Detection via Low-level Hardware Information
C. Woralert, C. Liu, ***Z. Blasingame****, and Z. Yang*
Published at AsianHOST (2023) · *Oral*

AsianHOST — 2022
Best Paper Nominee

HARD-Lite: A Lightweight Hardware Anomaly Realtime Detection Framework Targeting Ransomware
C. Woralert, C. Liu, and ***Z. Blasingame***
Published at AsianHOST (2022) · *Best Paper Nominee*

IJCB — 2021
Oral

Leveraging Adversarial Learning for the Detection of Morphing Attacks
Z. Blasingame *and C. Liu*
Published at IJCB (2021) · *Oral*
slides

ICANN — 2021
Oral

Feature Creation Towards the Detection of Non-control-Flow Hijacking Attacks
Z. Blasingame*, C. Liu, and X. Yao*
Published at ICANN (2021) · *Oral*
slides

HASP — 2019

Detecting Non-Control-Flow Hijacking Attacks Using Contextual Execution Information
G. Torres, Z. Yang, ***Z. Blasingame****, J. Bruska, and C. Liu*
Published at HASP (2019)

RESEC — 2018

Detecting Data Exploits Using Low-Level Hardware Information: A Short Time Series Approach
C. Liu, Z. Yang, ***Z. Blasingame****, G. Torres, and J. Bruska*
Published at RESEC (2018)

SPIE — 2017

Verification of OpenSSL Version via Hardware Performance Counters
J. Bruska, ***Z. Blasingame****, and C. Liu*
Published at SPIE (2017)

Invited Talks

AITHYRA *Vienna, Austria*	**Strong Stochastic Flow Maps** *AI x Bio Seminar. Organized by Michael Bronstein and Georg Winter* slides	07.2026
MolSS *Online*	**Strong Stochastic Flow Maps** *Machine Learning for Molecule Simulations & Sampler Reading Group* slides	07.2026
Starkly Speaking *Online*	**Strong Stochastic Flow Maps** *Starkly Speaking Reading Group. Organized by Hannes Stärk* slides video	07.2026
Mila *Montréal, Canada*	**AdjointDEIS: Efficient Gradients for Diffusion Models** *Sampling Reading Group. Organzied by Kirill Neklyudov* slides	01.2025
Embassy of the Kingdom of the Netherlands *Washington, D.C.*	**Diffusion Morphs (DiM): Diffusion is all you need for highly effective face morphs** *Transatlantic Dialogue on Presentation Attack Detection* slides	11.2024
IEEE IJCB *Buffalo, NY*	**Diffusion Morphs (DiM): Leveraging Diffusion for Strong and High-Quality Face Morphing Attacks** *Journal Presentation* slides	09.2024
Idiap *Martigny, Switzerland*	**Diffusion Morphs (DiM): The Power of Iterative Generative Models for Attacking FR Systems** *Biometrics Security & Privacy Group. Organized by Sébastien Marcel* slides	07.2024
DSA *Online*	**Diffusion for the Generation of Face Morphs** *CITeR & Document Security Alliance* slides	02.2024

Awards and Honors

NeurIPS Oral Presentation [top 0.4% of submissions]	2026
NeurIPS Top Reviewer Award	2026
ICML Oral Presentation [top 0.7% of submissions]	2026
ICML Gold Reviewer Award	2026
NeurIPS Top Reviewer Award	2025
AISTATS Best Reviewer Award	2025
IJCB Doctoral Consortium	2024
Best Oral Presentation for Software Engineering — Clarkson RAPS	2017
Clarkson Presidential Scholar	2015 – 2018

Teaching

Griffiss Institute *Rome, NY*	**Instructor** • Cyber Summer Camp • Advanced Cyber Summer Camp	2020 – 2024
Clarkson University *Potsdam, NY*	**Graduate Teaching Assistant** • Electrical Science • Digital Design • ECE Sophomore Lab	2018 – 2021
Clarkson University *Potsdam, NY*	**Undergraduate Teaching Assistant** • Software System Architecture • Differential Equations	2016 – 2018

Funding

CITeR	**Explainable Image Quality with Transformer-based Models** *NSF Industry–University Cooperative Research Center. Primary student investigator. PI: Chen Liu*	2023 – 2024
CITeR	**Towards the Creation of a Large Dataset of High-Quality Face Morphs** *NSF IUCRC. Primary student investigator at Clarkson. PIs: Chen Liu, Stephanie Schuckers, Xin Li, Jeremy Dawson, Nasser Nasrabadi, David Doermann, Srirangaraj Setlur, Siwei Lyu, Xiaoming Liu, Sébastian Marcel*	2021 – 2024
CITeR	**Comparative Detection of Facial Image Manipulation Techniques** *NSF IUCRC. Primary student investigator. PI: Chen Liu*	2020 – 2022
CITeR	**Adversarial Learning Based Approach Against Face Morphing Attacks** *NSF IUCRC. Primary student investigator. PI: Chen Liu*	2019 – 2021

Professional Services

Area Chair	NeurIPS Workshop on `AI for Stochastic Dynamics` (2026)
Reviewer at NeurIPS	2024, 2025 (Top Reviewer), 2026 (Top Reviewer)
Reviewer at ICML	2025, 2026 (Gold Reviewer)
Reviewer at ICLR	2026
Reviewer at AISTATS	2025 (Best Reviewer)
Workshop Reviewer	NeurIPS Frontiers in Probabilistic Inference (2025); ICCV `2COOOL` (2025); ICLR `Deep Generative Models` (2025); ICLR `Frontiers in Probabilistic Inference` (2025); IJCB `Face Morphing Attack and Detection Techniques` (2024)
Other Reviewing	IEEE ICPR (2024); IEEE MOST (2023–2024); IEEE BTAS (2019)

Skills

ML Architectures	Diffusion Models, Flow Matching, Neural Differential Equations, GANs, VAEs
Distributed Computing	Slurm, CUDA, ROCm, Linux

Languages	Python, C/C++, Java, Javascript, Bash, MATLAB, VHDL
Frameworks	TensorFlow, Pytorch, Numpy, Scipy, Matplotlib, Plotly, Pandas, Jax
Tools	Vim, Git, Docker, LaTeX
Audio	Over a decade of experience as a FOH engineer including theater productions and worship services

References

References available upon request